Uncertainty-Aware Multi-Agent Ransomware Detection via Decision MLP Fusion, Score-Space Augmentation, and Bayesian Triage Policy Optimisation

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Abstract—Ransomware detection demands more than a binary verdict: it requires uncertainty estimation, heterogeneous evidence fusion, and uncertainty-driven triage. We present a three-agent framework with four contributions. First, a Semantic CNN (static PE embeddings), a Behavioural CNN (16D hypervisor telemetry), and a Knowledge CNN (ATT&CK tactic embeddings) each use Monte Carlo Dropout for predictive probability and epistemic uncertainty. Second, a Decision MLP learns a temperature-calibrated risk score (ECE 0.026) from the six-dimensional score pool; since the Semantic and Knowledge Agents contribute only a fixed corpus-mean score during RanSMAP fusion, naive mean fusion is well-calibrated (ECE 0.023), so fusion's value lies in the uncertainty structure it exposes for triage rather than a calibration or AUROC gain; CTGAN augmentation addresses score-space imbalance without retraining base agents. Third, a four-threshold triage policy optimised via Bayesian TPE cuts escalation from 54.3% to 29.1% while raising balanced accuracy from 0.770 to 0.865. Fourth, Peer Consultation mitigates knowledge collapse under concurrent execution, and an Expert Feedback Loop with Elastic Weight Consolidation limits catastrophic forgetting (ΔAUPRC=0.001). Over six seeds and four generalisation splits on RanSMAP, RDset, and RansKnow, the framework achieves AUROC 0.996 (static PE) and 0.963 (behavioural), matched by the fused Decision MLP (0.952). Family holdout AUROC falls to 0.707 (0.677 fused): epistemic uncertainty routes most novel-family samples to escalation rather than silent miss, though Ryuk is an exception where the system is confidently wrong. A Mondrian conformal prediction wrapper adds distribution-free coverage under exchangeability, auto-classifying 54% of traffic at 99% coverage.

Index Terms—Ransomware detection, multi-agent systems, uncertainty quantification, generative adversarial networks, continual learning, conformal prediction.

I. Introduction

Ransomware attacks impose escalating operational and financial costs on enterprises and critical infrastructure [1], [2]. State-of-the-art detectors—signature-based antivirus, EDR platforms, or machine-learning classifiers on file-level features—converge on a common architectural limitation: a single scalar confidence score mapped to a binary verdict. When a novel variant or obfuscated execution degrades that score silently, no signal informs the operator that the verdict is unreliable.

We identify three structural gaps in the current state of practice:

1) No principled uncertainty. A softmax output from a deterministic neural network is a point estimate of class membership, not a probability over models. It cannot distinguish a confident correct prediction from a confident incorrect one.
2) Heuristic modality fusion. Combining static PE features with runtime telemetry via equal-weight averaging discards the per-modality reliability signal; a semantic agent operating on encrypted PE sections and a behavioural agent observing obfuscated I/O patterns should not carry equal weight.
3) Binary triage. An alert generated by every uncertain detection overwhelms analysts; equally, routing every uncertain detection to automatic containment risks business disruption.

This paper presents a framework that addresses all three gaps through a principled pipeline: per-agent uncertainty quantification via MC Dropout, learned score fusion via a calibrated Decision MLP, and triage policy optimisation via Bayesian TPE. The framework treats uncertainty as a control variable for deployment rather than a diagnostic afterthought.

Contributions

1) A heterogeneous three-agent architecture in which a Semantic CNN, a Behavioural CNN, and a Knowledge CNN each produce a predictive probability and an epistemic uncertainty via $T = 10$ MC Dropout forward passes.
2) A Decision MLP fusion mechanism with temperature calibration (ECE 0.026$\pm$0.003) that implicitly down-weights the Knowledge Agent under elevated uncertainty; because the Semantic and Knowledge Agents contribute a fixed corpus-mean score for RanSMAP samples, naive mean fusion over the same pool is itself well-calibrated (ECE 0.023), so its principal value lies in the uncertainty structure it exposes for the triage policy (Section IX-G) rather than

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a calibration or AUROC advantage, supported by CTGAN score-space augmentation to handle class imbalance at the fusion layer.
3) A formal triage policy over $(\hat{p}, \sigma_{\max})$ space with four jointly optimised thresholds via Bayesian TPE, reducing escalation rate from 54.3% to 29.1% while improving balanced accuracy from 0.770 to 0.865.
4) A Peer Consultation mechanism that prevents knowledge collapse under concurrent execution, and an Expert Feedback Loop with EWC enabling continual family adaptation without catastrophic forgetting.

II. Related Work

A. Behavioural Ransomware Detection

Hypervisor-level behavioural monitoring [3], [4] captures low-level I/O and memory access patterns before encryption artefacts surface in the filesystem, providing strong coverage of novel variants, though IRP monitors [5] remain vulnerable to slow-encrypt evasion and API-call classifiers [6] require costly process-level instrumentation. Survey work [7] confirms that runtime behavioural methods outperform static analysis against obfuscated ransomware, but notes a structural limitation shared by all existing systems: they produce a scalar confidence score with no mechanism for expressing per-sample epistemic uncertainty or routing uncertain detections to analyst escalation.

B. Uncertainty Quantification in Detection

Monte Carlo Dropout [8] re-activates dropout at test time, treating each stochastic forward pass as a draw from an approximate posterior over network weights, with the variance of T outputs approximating epistemic uncertainty at $T\times$ the inference cost. Gawlikowski et al. [9] establish MC Dropout and deep ensembles [10] as the most broadly applicable of four neural-network uncertainty method families, though ensembles require retaining M full model copies at inference. For distribution-free guarantees, conformal prediction [11] and its Mondrian variant [12] bound coverage probability without parametric assumptions, generalised to arbitrary risk functions by Angelopoulos and Bates [13]. Uncertainty estimation has been applied to network anomaly detection [14], but ransomware triage—where an autonomous-containment error differs qualitatively in cost from an escalation—has not previously been framed as an uncertainty-conditioned routing problem.

C. Multi-Agent and Ensemble Detection

Ensemble methods for malware classification [15], [16] aggregate multiple classifiers over a shared feature space via majority vote or average-score fusion. Multi-agent intrusion detection systems [17] distribute detection across specialised agents monitoring different network zones, combined through a centralised coordinator. Multi-modal ransomware detection combining static PE features, API-call sequences, and file-system events [7] typically uses equal-weight or attention-based fusion that does not propagate per-modality reliability.

Our framework differs structurally in two respects: the three agents operate on structurally heterogeneous input spaces ($\mathbb{R}^{384}$ PE embeddings, $\mathbb{R}^{16}$ runtime telemetry, $\mathbb{R}^{23}$ ATT&CK vectors) that cannot be concatenated without discarding modality-specific structure, and each agent contributes not only a probability but an epistemic uncertainty, which the Decision MLP uses implicitly to down-weight unreliable modalities and which the orchestrator uses explicitly to route uncertain cases to analyst escalation—a mechanism absent from standard ensemble methods.

D. Generative Augmentation and Continual Learning

Class imbalance in detection datasets is routinely addressed by SMOTE [18] and Conditional Tabular GAN (CTGAN) [19], applied to fraud and intrusion detection [20], [21] and consistently outperforming SMOTE when the minority class is small and non-Gaussian. We apply CTGAN at the score space output of the multi-agent detector: operating on bounded $[0, 1]^6$ probability–uncertainty vectors constrains synthetic samples to the agents' output range, reducing the risk of implausible combinations and eliminating raw-telemetry reconstruction artefacts.

Catastrophic forgetting—gradient updates for a new task systematically overwriting prior weight configurations [22]—is the key barrier to continual adaptation; temporal evaluation frameworks [23], [24] establish that standard classifiers lose substantial accuracy within months of the training cutoff under concept drift. Elastic Weight Consolidation (EWC) [22] mitigates this via a diagonal Fisher penalty protecting parameters important to retained tasks; our Expert Feedback Loop applies EWC to the Decision MLP, achieving ΔAUPRC=0.001 on retained families after five correction iterations—enabling continual adaptation without retraining base agents.

E. Relationship to Prior Work

This paper extends the two-agent Agentic SABRE framework [25] (Semantic + Behavioural CNN, MC Dropout, fixed triage policy) with four technically distinct contributions: (1) a third Knowledge CNN agent on ATT&CK tactic embeddings, expanding the score pool from $[0, 1]^4$ to $[0, 1]^6$; (2) Bayesian TPE optimisation of four jointly tuned triage thresholds $(\tau_{\text{high}}, \tau, \kappa, \kappa_{\text{low}})$ replacing the fixed (τ=0.10, κ=0.10) pair; (3) CTGAN score-space augmentation at the fusion layer; and (4) EWC continual learning and Mondrian conformal prediction wrappers. All experiments and reported numerical results in this paper were rerun independently; no numerical results or figures are reused from [25], though both papers draw on the same underlying RDset and RanSMAP datasets.

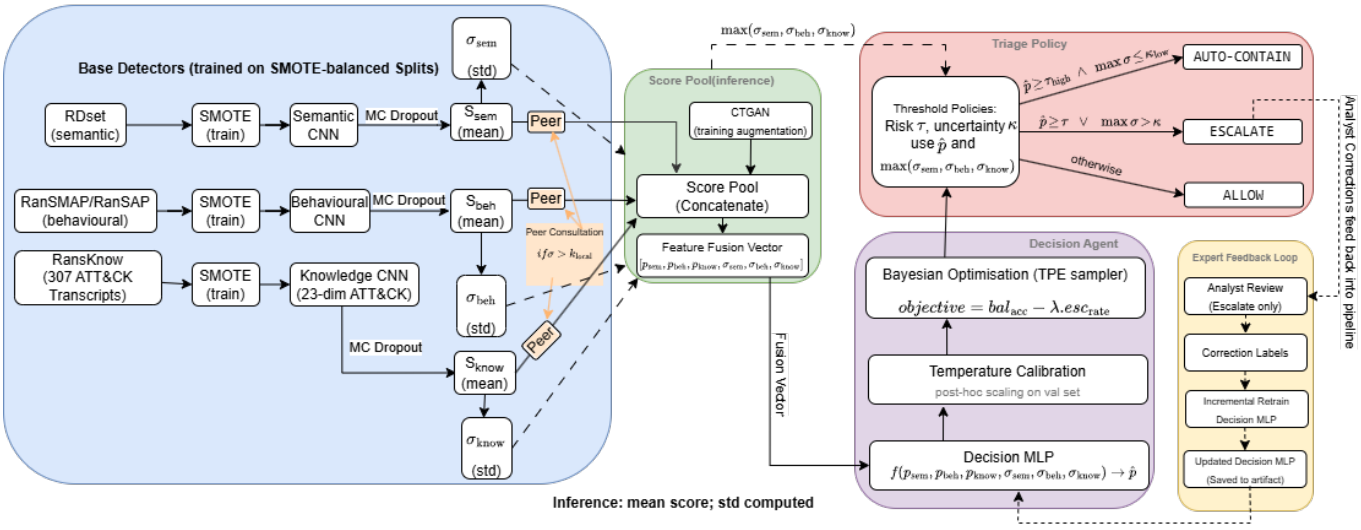


Fig. 1. End-to-end framework pipeline. Three specialised CNN agents (Semantic, Behavioural, Knowledge) each run T=10 MC Dropout forward passes to produce a probability–uncertainty pair $(\hat{p}_i, \sigma_i)$. Peer Consultation adjusts any agent whose σ_i exceeds its local threshold before the six-dimensional score pool is assembled. The Decision MLP fuses the pool into a calibrated risk score; Bayesian TPE derives the four-threshold triage policy that maps $(\hat{p}, \sigma_{\max})$ to Auto-Contain, Escalate, or Allow. Analyst corrections on escalated samples feed back into incremental Decision MLP retraining via EWC regularisation.

III. Problem Formulation

A. Detection Task

Let $\mathcal{X} = \mathcal{X}_{\text{sem}} \times \mathcal{X}_{\text{beh}} \times \mathcal{X}_{\text{know}}$ denote the joint input space of three heterogeneous modalities, and let $\mathcal{Y} = \{0, 1\}$ denote the binary ransomware label. A sample is observed as a triple $\mathbf{x} = (\mathbf{x}_{\text{sem}}, \mathbf{x}_{\text{beh}}, \mathbf{x}_{\text{know}}) \in \mathcal{X}$, where

- $\mathbf{x}_{\text{sem}} \in \mathbb{R}^{384}$ is a sentence-transformer embedding of Portable Executable (PE) section names;
- $\mathbf{x}_{\text{beh}} \in \mathbb{R}^{16}$ is a 16-dimensional feature vector extracted from a $T_{\text{window}} = 10\,\text{s}$ telemetry window (see Section IV);
- $\mathbf{x}_{\text{know}} \in \mathbb{R}^{d}$ is an embedding of ATT&CK tactic profiles derived from threat-intelligence transcripts.

B. Triage Task

Rather than mapping $\mathbf{x}$ to a binary label, the triage task maps $\mathbf{x}$ to one of three actions $\mathcal{A} = \{\text{Auto-Contain}, \text{Escalate}, \text{Allow}\}$ according to a policy $\pi : (\hat{p}, \sigma_{\max}) \to \mathcal{A}$, where $\hat{p} \in [0, 1]$ is the fused risk score and $\sigma_{\max} \in [0, 1]$ is the maximum per-agent epistemic uncertainty.

Definition 1 (Triage Policy). Given thresholds $\Theta = (\tau_{\text{high}}, \tau, \kappa, \kappa_{\text{low}})$ with $0 < \tau \le \tau_{\text{high}} \le 1$ and $0 < \kappa_{\text{low}} \le \kappa \le 1$, the triage policy is:

$$\pi(\hat{p}, \sigma_{\max}; \Theta) = \begin{cases} \text{Auto-Contain} & \hat{p} \ge \tau_{\text{high}} \ \wedge \ \sigma_{\max} \le \kappa_{\text{low}}, \\ \text{Escalate} & \hat{p} \ge \tau \ \vee \ \sigma_{\max} > \kappa, \\ \text{Allow} & \text{otherwise.} \end{cases} \tag{1}$$

Since $\tau \le \tau_{\text{high}}$, the Auto-Contain and Escalate conditions overlap on the region $\hat{p} \ge \tau_{\text{high}} \wedge \sigma_{\max} \le \kappa_{\text{low}}$: this is not an ambiguity but a stated precedence order, evaluated top-to-bottom, so Auto-Contain takes priority whenever both conditions hold.

C. Optimisation Objective

Let bal_acc(Θ) denote the balanced accuracy of the joint detection–triage system under thresholds Θ, and let esc_rate(Θ) denote the fraction of samples routed to Escalate. The policy optimisation problem is:

$$\Theta^* = \arg\max_{\Theta \in \mathcal{F}} \ J(\Theta) := \text{bal_acc}(\Theta) - \lambda\, \text{esc_rate}(\Theta), \tag{2}$$

where $\lambda \ge 0$ is an operational cost parameter encoding the relative burden of analyst escalation, and $\mathcal{F}$ is the feasibility region $\{\Theta : \tau \le \tau_{\text{high}},\ \kappa_{\text{low}} \le \kappa\}$.

Remark 1. The parameter λ is a deployment decision, not a modelling one. An organisation with constrained analyst capacity sets higher λ; one prioritising missed-detection avoidance sets $\lambda \approx 0$.

IV. Agent Architecture and Uncertainty Quantification

A. Monte Carlo Dropout as Approximate Bayesian Inference

Let $f_\theta : \mathcal{X}_i \to [0, 1]$ denote a neural network with parameters θ trained on modality i; standard dropout is disabled at inference, yielding a deterministic point estimate. MC Dropout [8] instead keeps dropout active at test time, treating each stochastic forward pass as a draw from an approximate posterior $q_\phi(\theta) \approx p(\theta \mid \mathcal{D})$.

Formally, for T stochastic forward passes with dropout masks $\{\mathbf{z}_t\}_{t=1}^{T}$:

$$\hat{p}_i(\mathbf{x}) = \frac{1}{T}\sum_{t=1}^{T} f_{\theta\odot\mathbf{z}_t}(\mathbf{x}), \tag{3}$$

$$\sigma_i(\mathbf{x}) = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}\left(f_{\theta\odot\mathbf{z}_t}(\mathbf{x}) - \hat{p}_i(\mathbf{x})\right)^2}, \tag{4}$$

where $\odot$ denotes element-wise masking. The quantity σ_i captures epistemic uncertainty—reducible uncertainty arising from limited training data—as distinct from aleatoric uncertainty inherent to the data generating process [26].

Proposition 1 (Epistemic Signal). When a test sample $\mathbf{x}$ lies far from the training manifold of agent i (i.e., a novel ransomware family), we hypothesise that the variance $\sigma_i^2(\mathbf{x})$ tends to increase with distributional shift $d(\mathbf{x}, \mathcal{D}_{\text{train}})$ [8]. Approximate Bayesian methods do not provide a formal monotonicity guarantee in general; we verify this tendency empirically across family and hardware holdout splits (Supplementary Section S-VI).

We use $T = 10$ passes per agent; a T-sweep against a near-converged T=200 reference (Section IX-O) shows σ_{beh} reaches within 11% of that reference by T=10, down from 100% at T=1, at 1.9 ms latency versus 9.5 ms at T=50 (which itself is only within 2% of the reference). Supplementary Section S-VI quantifies this stability and the associated latency trade-off.

B. Semantic Agent

The Semantic Agent operates on static PE structure. Given a PE file, section names are concatenated and encoded via a frozen sentence transformer (all-MiniLM-L6-v2) to produce $\mathbf{x}_{\text{sem}} \in \mathbb{R}^{384}$. A Semantic CNN f_{sem} with MC Dropout maps this embedding to $(\hat{p}_{\text{sem}}, \sigma_{\text{sem}})$.

Training data: RDset [27], [28], comprising 138,047 PE files (41,123 ransomware; 96,924 benign) with SMOTE balancing applied at the feature level prior to training.

C. Behavioural Agent

The Behavioural Agent operates on runtime storage and memory access patterns collected via a thin BitVisor hypervisor [4]. Within each $T_{\text{window}} = 10$ s window we compute a $\mathbf{x}_{\text{beh}} \in \mathbb{R}^{16}$ feature vector partitioned as $\mathbf{x}_{\text{beh}} = (\chi_s, \chi_m)$, where:

Storage features $\chi_s \in \mathbb{R}^5$:

$$\chi_s = \left(\bar{T}_{s,r},\ \bar{T}_{s,w},\ V_{s,r},\ V_{s,w},\ \bar{H}_{s,w}\right), \tag{5}$$

where $\bar{T}_{s,r}$ and $\bar{T}_{s,w}$ are mean read/write throughputs [byte/s]; $V_{s,r}$ and $V_{s,w}$ are variances of accessed Logical Block Addresses (LBA); and $\bar{H}_{s,w}$ is mean Shannon entropy of written sectors, computed as $H(X) = -\sum_i p(x_i)\log_2 p(x_i)$.

Memory features $\chi_m \in \mathbb{R}^{11}$:

$$\chi_m = \Big(H_{m,w},\ H_{m,rw},\ C_{4K,r},\ C_{4K,w},\ C_{4K,rw},\ C_{4K,x},\ C_{2M,r},\ C_{2M,rw},\ C_{\text{MMIO},r},\ C_{\text{MMIO},w},\ V_{m,w}\Big), \tag{6}$$

where $H_{m,w}$ and $H_{m,rw}$ are Shannon entropies of written and read/written memory pages; $C_{s,o}$ denotes the count of memory accesses of size $s \in \{4\,\text{KiB}, 2\,\text{MiB}, \text{MMIO}\}$ and operation $o \in \{r, w, rw, x\}$; and $V_{m,w}$ is the variance of written Guest Physical Addresses (GPA).

Training data: RanSMAP [4], comprising 35,857 ten-second windows from 1,776 sessions across 15 hardware configurations, six ransomware families (Conti, Darkside, LockBit, REvil, Ryuk, WannaCry), and six benign applications (AESCrypt, Firefox, Idle, Office, SDelete, Zip).

D. Knowledge Agent

The Knowledge Agent encodes ATT&CK tactic profiles derived from threat-intelligence transcripts (RansKnow [29]: 718 ATT&CK-aligned transcripts). A Discourse CNN f_{know} maps the tactic embedding $\mathbf{x}_{\text{know}} \in \mathbb{R}^d$ to $(\hat{p}_{\text{know}}, \sigma_{\text{know}})$.

E. Evaluation Protocol and Generalisation Splits

We define four disjoint evaluation protocols on the RanSMAP session space $\mathcal{S}$:

1) Random: 80/20 stratified split over $\mathcal{S}$.
2) Family holdout (more precisely a class holdout: the held-out set spans both ransomware families and benign applications): $\mathcal{S}_{\text{test}} = \{s \in \mathcal{S} : \text{family}(s) \in \{\text{REvil}, \text{WannaCry}, \text{Firefox}, \text{Zip}\}\}$. We retain the shorter name "family holdout" throughout for brevity.
3) Hardware holdout: $\mathcal{S}_{\text{test}} = \{s \in \mathcal{S} : \text{hw}(s) \in \{\text{i5-gen12}, \text{i7-gen12}, \text{i3-gen12/DDR5}\}\}$.
4) Temporal: $\mathcal{S}_{\text{test}} = \{s \in \mathcal{S} : t(s) \geq t_{0.85}\}$, where $t_{0.85}$ is the 85th percentile timestamp.

All results are reported as mean ± std over $K = 6$ independent random seeds.

V. Decision MLP Fusion and Score-Space Augmentation

A. Score Pool Construction

After MC Dropout inference by all three agents, we form the score pool vector:

$$\mathbf{v}(\mathbf{x}) = \left(\hat{p}_{\text{sem}},\ \sigma_{\text{sem}},\ \hat{p}_{\text{beh}},\ \sigma_{\text{beh}},\ \hat{p}_{\text{know}},\ \sigma_{\text{know}}\right) \in [0,1]^6. \tag{7}$$

The maximum per-agent uncertainty $\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}})$ is derived from $\mathbf{v}$ and used independently in the triage policy (Section VI), but is not fed to the Decision MLP.

Remark 2 (Cross-Dataset Fusion Without Per-Sample Pairing). RDset, RanSMAP, and RansKnow are independently collected: no executable appears in more than one dataset, so a RanSMAP execution window has no corresponding PE binary or ATT&CK transcript. We therefore do not attempt to condition $\hat{p}_{\text{sem}}$ or $\hat{p}_{\text{know}}$ on the RanSMAP sample's own (family) identity—doing so via a per-family lookup keyed by the sample's true label would leak that label into the score pool. Instead, for every RanSMAP evaluation, $(\hat{p}_{\text{sem}}, \sigma_{\text{sem}})$ and $(\hat{p}_{\text{know}}, \sigma_{\text{know}})$ are

TABLE I
Which dataset each agent is trained and evaluated on, and what it contributes to a RanSMAP fusion score $\mathbf{v}(\mathbf{x})$ (Remark 2). Only the Behavioural Agent's contribution to $\mathbf{v}$ varies per RanSMAP sample.

Agent	Trained & evaluated on	Contribution to RanSMAP $\mathbf{v}(\mathbf{x})$
Semantic	RDset (own train/test split)	Fixed corpus-mean $(\hat{p}_{\text{sem}}, \sigma_{\text{sem}})$, identical $\forall\, \mathbf{x}$
Behavioural	RanSMAP (own train/test split)	Per-sample $(\hat{p}_{\text{beh}}(\mathbf{x}), \sigma_{\text{beh}}(\mathbf{x}))$
Knowledge	RansKnow (own train/test split)	Fixed corpus-mean $(\hat{p}_{\text{know}}, \sigma_{\text{know}})$, identical $\forall\, \mathbf{x}$

fixed, label-independent constants: the MC Dropout mean and standard deviation of each agent's own corpus (RDset for Semantic, RansKnow for Knowledge), computed once and reused for every RanSMAP sample regardless of family. Consequently, in the RanSMAP evaluations of Section IX, only the Behavioural Agent carries per-sample discriminative signal; the Semantic and Knowledge Agents contribute a fixed prior that the Decision MLP can learn to weight, and Peer Consultation (Section VII) can still substitute when σ_{know} or σ_{sem} exceeds a peer's, but neither agent can improve fused AUROC over the Behavioural Agent alone in this setting. This is why the Decision MLP's contribution below is measured in calibration and uncertainty-gated triage quality rather than in raw discrimination. Table I makes this mapping explicit.

The Semantic and Knowledge Agents' own in-domain AUROC (RDset 0.996; Knowledge Agent metrics not separately reported given RansKnow's small size, Section X) therefore says nothing about their contribution to a RanSMAP fusion score, which is fixed and sample-independent by construction; the Semantic Agent's genuine sample-level contribution is validated separately, in-domain, on RDset.

Remark 3 (Why Score Space). Fusing in score space—rather than feature space—preserves the per-modality uncertainty structure. Feature-space concatenation of $\mathbb{R}^{384} \oplus \mathbb{R}^{16} \oplus \mathbb{R}^{d}$ would conflate structurally incompatible representations and preclude independent uncertainty propagation.

B. CTGAN Score-Space Augmentation

Let $\mathcal{V}_{\text{train}} = \{(\mathbf{v}_j, y_j)\}_{j=1}^{N}$ denote the score-pool training set with labels $y_j \in \{0, 1\}$, where the ransomware class is a fraction $r \ll 0.5$ of samples. We apply Conditional Tabular GAN (CTGAN) [19] to generate synthetic minority-class score-pool vectors.

The CTGAN minimax objective is:

$$\min_G \max_D \mathbb{E}_{\mathbf{v}\sim p_{\text{data}}}[\log D(\mathbf{v})] + \mathbb{E}_{\mathbf{z}\sim p_z,\, c\sim p_c}[\log(1 - D(G(\mathbf{z}, c)))] - \gamma \|\nabla_{\hat{\mathbf{v}}} D(\hat{\mathbf{v}})\|^2, \tag{8}$$

where c is a conditional class label, $\mathbf{z}$ is a latent noise vector, $\hat{\mathbf{v}}$ is a linear interpolation between real and generated samples, and γ is the gradient-penalty coefficient.

Crucially, CTGAN operates on $\mathbf{v} \in [0, 1]^6$, not on raw features: this ensures that the generator cannot produce physically impossible score combinations and preserves the calibration structure of the base agents.

C. Decision MLP Architecture

The Decision MLP $g_\phi : [0, 1]^6 \to [0, 1]$ maps the score pool to a fused risk score:

$$g_\phi(\mathbf{v}) = \sigma(\mathbf{W}_2\, \text{ReLU}(\mathbf{W}_1 \mathbf{v} + \mathbf{b}_1) + \mathbf{b}_2)\,, \tag{9}$$

with dimensions $6 \to 32 \to 1$ and sigmoid output. Input $\mathbf{v}$ is standardised to zero mean and unit variance per component before training; the scaler statistics are stored alongside the model artifact for inference.

D. Temperature Calibration

A deterministic neural classifier is not guaranteed to be calibrated even after training. We apply temperature scaling [30]: the logit $\ell = \log(\hat{p}/(1 - \hat{p}))$ is divided by a learned scalar $T > 0$ before the final sigmoid:

$$\hat{p}_{\text{cal}} = \sigma(\ell / T), \quad T^* = \arg\min_T \text{NLL}_T(\mathcal{D}_{\text{cal}}). \tag{10}$$

Expected Calibration Error is defined as:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\text{acc}(B_m) - \text{conf}(B_m)\right|, \tag{11}$$

where $\{B_m\}_{m=1}^{M}$ partition $[0, 1]$ into M equal-width confidence bins, $\text{acc}(B_m)$ is empirical accuracy and $\text{conf}(B_m)$ is mean predicted confidence within bin B_m.

The mean fusion baseline (ECE = 0.023) and the Decision MLP with temperature scaling (ECE = 0.026) are essentially tied (Section IX-E).

VI. Triage Policy and Bayesian Optimisation

A. Triage Policy in $(\hat{p}, \sigma_{\max})$ Space

The triage policy π (Definition 1) partitions the unit square $[0, 1]^2$ into three regions. The fused risk $\hat{p} = g_\phi(\mathbf{v})$ and maximum agent uncertainty $\sigma_{\max}$ jointly determine the action. The boundary conditions ensure that:

- Auto-Contain requires both high risk and low uncertainty—autonomous action is taken only when the system is jointly confident in detection and reliable;
- Escalate fires when either the risk score is elevated ($\hat{p} \geq \tau$) or agent disagreement is high ($\sigma_{\max} > \kappa$), routing borderline and uncertain samples to an analyst rather than acting autonomously;
- Allow covers the confident-benign region.

Fig. 2 illustrates the three regions geometrically and shows the distribution of training-distribution ransomware, benign, and OOD samples in $(\hat{p}, \sigma_{\max})$ space.

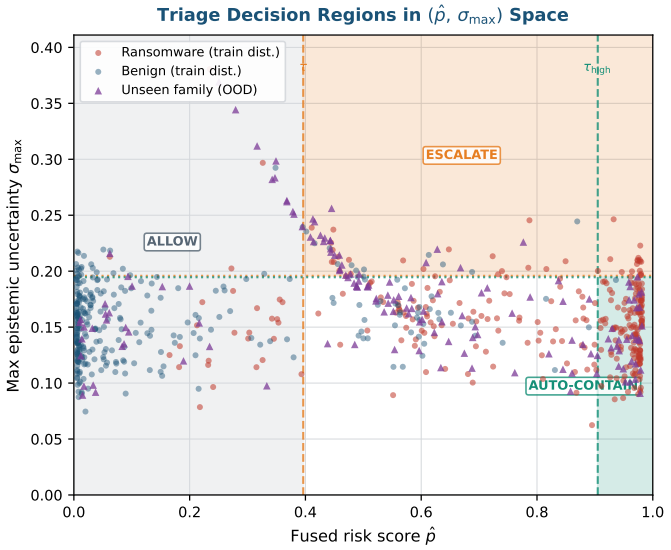


Fig. 2. Triage decision regions in $(\hat{p}, \sigma_{\max})$ space under the real Bayesian TPE-optimised thresholds (τ_{high}=0.905, τ=0.396, κ=0.196, κ_{low}=0.195) from checkpoints/policy_auto.json. Points are genuine fused [S+B+K] Decision MLP output: ransomware and benign from the random-stratified test split (in-distribution for that checkpoint), and OOD unseen-family samples (△) from the family-holdout test split restricted to REvil and WannaCry, the two families that checkpoint never trained on. OOD samples concentrate in the Escalate region (63%) due to elevated $\sigma_{\max}$, though not exclusively (19% still reach Auto-Contain).

B. Bayesian Triage Policy Optimisation via TPE

The objective $J(\Theta)$ in (2) is non-differentiable in Θ (discrete triage counts), so we employ Bayesian optimisation via Tree-structured Parzen Estimation (TPE) [31].

TPE models the probability density of J conditioned on Θ as:

$$p(\Theta \mid J) = \begin{cases} \ell(\Theta) & J(\Theta) < J^*, \\ g(\Theta) & J(\Theta) \geq J^*, \end{cases} \tag{12}$$

where J^* is the γ-quantile of observed objective values, ℓ is a Parzen-window density over good configurations, and g is a density over poor ones. The acquisition criterion selects:

$$\Theta_{\text{next}} = \arg\max_{\Theta} \frac{\ell(\Theta)}{g(\Theta)}, \tag{13}$$

which approximates the Expected Improvement criterion.

We run 300 TPE trials over $\mathcal{F}$ with $\lambda = 0.3$; the search budget is chosen based on the dimensionality of Θ and the cost of evaluating J on the validation set (approximately 0.15 s per trial). Results and convergence analysis are presented in Section IX-G.

C. Comparison with Grid Search and Fixed Thresholds

Table II summarises the triage outcome under the three policies on the random-stratified test split. Following the RansKnow correction, Bayesian TPE and grid search converge to statistically indistinguishable balanced accuracy

TABLE II
Triage policy comparison on random-stratified test split (mean ± 95% CI over K=6 seeds). Bal. acc = balanced accuracy; Esc = escalation rate; AC = auto-contain rate. ↑ = higher is better; ↓ = lower is better.

Policy	Bal. acc ↑	Esc (%) ↓	AC (%) ↑
SABRE+ Fixed [25]	0.770 ± 0.001	54.3 ± 0.7	18.0 ± 0.6
Grid search	0.865 ± 0.001	30.2 ± 0.3	28.2 ± 0.3
Bayesian TPE (300 trials)	$\mathbf{0.865 \pm 0.001}$	$\mathbf{29.1 \pm 0.5}$	$\mathbf{29.3 \pm 0.5}$

(0.865 for both), with TPE retaining a modest edge in escalation rate (29.1% vs 30.2%) and auto-contain rate (29.3% vs 28.2%): optimisation still pays off, but the margin over grid search has narrowed substantially now that the underlying Knowledge score is more reliable. Fixed thresholds (τ=0.10, κ=0.10, inherited from [25]) are no longer near-random: with the corrected Knowledge Agent they reach 0.770 balanced accuracy at 54.3% escalation—workable on its own, though still noticeably less efficient than either optimised policy, which improves balanced accuracy by 0.095, roughly halves escalation (54.3% → 29.1%), and increases the auto-contain rate by 1.6× (18.0% → 29.3%).

VII. Peer Consultation and Expert Feedback Loop

A. Peer Consultation

Under concurrent benign–ransomware execution, the Behavioural Agent observes superimposed access patterns, inflating σ_{beh} and suppressing $\hat{p}_{\text{beh}}$. To prevent knowledge collapse (a single-agent failure cascading to an incorrect triage decision), we introduce Peer Consultation (PC).

Definition 2 (Peer Consultation). When agent i's epistemic uncertainty $\sigma_i(\mathbf{x}) > \kappa_{\text{PC}}$ for a threshold $\kappa_{\text{PC}} > 0$, agent i's probability only is replaced by a weighted combination of its peer agents' current probabilities:

$$\hat{p}_i^{\text{PC}}(\mathbf{x}) = \sum_{j \neq i} w_j \, \hat{p}_j(\mathbf{x}), \tag{14}$$

where weights are proportional to the inverse uncertainty of each peer, $w_j \propto 1/\sigma_j$, normalised so that $\sum_{j \neq i} w_j = 1$. Agent i's own uncertainty σ_i is not modified by this substitution—it is used unchanged by the triage policy (Section VI), so a consulted agent still contributes its original (elevated) uncertainty to $\sigma_{\max}$. Each agent maintains its own threshold $\kappa_{\text{PC}}^{(i)}$; in practice all are set equal to the triage threshold κ optimised by Bayesian TPE.

The three triggering conditions are evaluated independently, in the fixed order Semantic → Behavioural → Knowledge, and each substitution uses whatever probabilities are current at that point in the sequence—so if two or three agents are simultaneously uncertain, a later agent's replacement can draw on an earlier agent's already-substituted probability rather than its original one. For example, if both σ_{sem} and σ_{beh} exceed κ_{PC}, the Semantic Agent's probability is substituted first (from

the Behavioural and Knowledge Agents' original probabilities), and the Behavioural Agent's substitution then draws on the Semantic Agent's already-substituted probability together with the Knowledge Agent's original one. Peer Consultation does not suppress the triggering agent's uncertainty; rather, it routes the sample to the Escalate action while ensuring the score pool reflects available evidence from the more reliable modalities.

B. Expert Feedback Loop with EWC

The Expert Feedback Loop allows an analyst-approved correction label $\tilde{y}$ for an escalated sample to update the Decision MLP g_ϕ incrementally via Elastic Weight Consolidation (EWC) [22], without catastrophic forgetting of the base agents' knowledge. Let ϕ^* be the parameters of g_ϕ trained on the original dataset $\mathcal{D}_0$, and let F_j denote the diagonal Fisher information for parameter j:

$$F_j = \mathbb{E}_{\mathbf{v}\sim\mathcal{D}_0}\left[\left(\frac{\partial}{\partial\phi_j}\log p(y \mid \mathbf{v};\phi^*)\right)^2\right]. \tag{15}$$

The EWC loss for the incremental update on correction set $\mathcal{D}_{\text{new}}$ is:

$$\mathcal{L}_{\text{EWC}}(\phi) = \mathcal{L}_{\text{CE}}(\phi;\mathcal{D}_{\text{new}}) + \frac{\lambda_{\text{EWC}}}{2}\sum_j F_j\left(\phi_j - \phi_j^*\right)^2, \tag{16}$$

where $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss on new samples and λ_{EWC} controls the trade-off between plasticity and stability. We measure catastrophic forgetting as:

$$\Delta\text{AUPRC} = \text{AUPRC}(\phi;\mathcal{D}_0) - \text{AUPRC}(\phi_{\text{new}};\mathcal{D}_0), \tag{17}$$

achieving $\Delta\text{AUPRC} = 0.001$ on retained families after five feedback iterations.

VIII. Conformal Prediction Wrapper

A. Mondrian Conformal Prediction

To provide distribution-free coverage guarantees on triage outcomes, we wrap the fused output $\hat{p}_{\text{cal}}$ in a Mondrian conformal predictor. Let $\alpha \in (0,1)$ be the target miscoverage rate.

Definition 3 (Conformal Prediction Set). Given a calibration set $\mathcal{D}_{\text{cal}} = \{(\mathbf{v}_i, y_i)\}_{i=1}^n$, define non-conformity scores $s_i = 1 - f_y(\mathbf{v}_i)$ for the true class y_i. The conformal prediction set at level α for a new sample $\mathbf{v}$ is:

$$\mathcal{C}_\alpha(\mathbf{v}) = \left\{y \in \mathcal{Y} : s(\mathbf{v},y) \le Q_{1-\alpha}(\{s_i\}_{i=1}^n)\right\}, \tag{18}$$

where $Q_{1-\alpha}$ is the $(1-\alpha)$-quantile of calibration scores.

Theorem 1 (Marginal Coverage Guarantee). Under exchangeability of calibration and test samples:

$$\mathbb{P}(Y \in \mathcal{C}_\alpha(\mathbf{v})) \ge 1 - \alpha. \tag{19}$$

Mondrian conformal prediction stratifies the calibration non-conformity scores by each calibration sample's true label (standard label-conditional Mondrian CP), conditioning (18) separately for each class. Theorem 1 provides a formal marginal coverage guarantee under exchangeability of calibration and test samples; Mondrian stratification yields class-conditional coverage empirically but does not carry a separate distribution-free guarantee beyond the marginal bound. The exchangeability assumption holds for the random-stratified split; under temporal and family holdout, the formal guarantee does not apply and coverage is verified empirically (Section IX-L).

IX. Experimental Evaluation

A. Datasets

RDset [27], [28]: 138,047 PE files (41,123 ransomware; 96,924 benign; ≈30% prevalence), used exclusively for Semantic Agent training/evaluation with disjoint train/test splits at the binary (file-hash) level.

RanSMAP [4]: 35,857 ten-second execution windows from 1,776 sessions across 15 hardware configurations (four CPUs: Celeron G3920, Core i3-12100, Core i5-12500, Core i7-12700; six DDR4/DDR5 frequencies; two RAM capacities), six ransomware families (Conti, Darkside, LockBit, REvil, Ryuk, WannaCry), and six benign applications (AESCrypt, Firefox, Idle, Office, SDelete, Zip) under BitVisor hypervisor surveillance; moderate class imbalance addressed via SMOTE at the feature level. A separate variants partition provides 1,397 windows from seven Conti sub-variants for OOD analysis; a mix partition provides 600 windows from ten concurrent execution combinations. Neither is used for training.

RansKnow [29]: 718 ATT&CK-aligned threat-intelligence transcripts, used exclusively for Knowledge Agent training. Score-space imbalance at the fusion layer is addressed separately by CTGAN augmentation (Section V).

B. Implementation Details

Agent architectures. All three base agents share one lightweight single-block design, scaled to their respective input dimensionality: a single-channel input sequence (of length 384, 16, or 23 respectively—the input dimensionality, not a channel count) through one Conv1d(1→C, k=3, same-padding)–ReLU–MaxPool1d(2)–Dropout(p=0.3) block, Flatten, FC(→ 64 or 128)–ReLU, FC(→ 1)–Sigmoid. The Semantic CNN takes $\mathbf{x}_{\text{sem}} \in \mathbb{R}^{384}$ with C=32 channels and a 128-unit hidden FC layer; SMOTE is applied to RDset at the embedding level (r_{minority}=0.30). The Behavioural CNN takes $\mathbf{x}_{\text{beh}} \in \mathbb{R}^{16}$ with C=16 channels and a 64-unit hidden FC layer (8,385 parameters, 32.8 KB); SMOTE is applied to RanSMAP training windows. The Knowledge CNN mirrors this architecture scaled to $\mathbf{x}_{\text{know}} \in \mathbb{R}^{23}$, trained on 718 RansKnow transcripts (70/15/15 train/val/test by family label). None of the three base agents use BatchNorm. The Decision MLP maps $\mathbf{v} \in [0,1]^6$ (standardised per component) through FC(6→32)–ReLU–Dropout(p=0.2), FC(32→1), Sigmoid, with temperature scaling applied post-hoc via L-BFGS on the validation set. Full architecture pseudocode is given in Supplementary Algorithms S1–S2.

Training and hyperparameters. All networks use Adam (η=10^{-3}, batch size 64, up to 50 epochs, early stopping patience 6, best weights restored), K=6 seeds for all RanSMAP experiments. CTGAN, Bayesian TPE, and EWC hyperparameters are given with their respective results in Sections IX-G and IX-M and Supplementary Algorithm S2 (ctgan v0.9, Optuna 3.5). Hardware: NVIDIA GB10 GPU, Python 3.10, PyTorch 2.2; inference latency measured over 500 repetitions (50 warm-up, batch size 32).

Out-of-fold Decision MLP training features. Scoring the Behavioural Agent's own training windows with a checkpoint trained on those same windows would let the Decision MLP fit an optimistically confident score it never sees again at test time. We therefore build the Behavioural component of every Decision MLP training vector out-of-fold: the random-stratified training sessions are partitioned into five session-grouped, stratified folds (StratifiedGroupKFold), a Behavioural CNN is trained on each set of four folds, and the held-out fifth fold is scored with that fold's own checkpoint; concatenating all five folds' held-out scores covers the full training set with no sample scored by a model that trained on it. The leave-one-family-out evaluation (Section IX-S) applies the same five-fold cross-fitting within each family-excluded training pool, so its Decision MLP training features are simultaneously family-excluded and out-of-fold; test-time scoring for each held-out family uses a Behavioural checkpoint trained on that family-excluded pool, never on the held-out family. Test-time inference throughout uses the full-training-set checkpoint on genuinely held-out test sessions and requires no cross-fitting.

C. Baseline Comparisons

We compare against:

1) Hirano and Kobayashi [4]: 1D-CNN + LSTM baseline on RanSMAP (23D features, $T_{\text{window}} = 0.1\,\text{s}$, $T_d = 30\,\text{s}$) — direct dataset comparison.
2) Mean fusion: [B+K] equal-weight average of $\hat{p}_{\text{beh}}, \hat{p}_{\text{know}}$ (Semantic excluded), without the Decision MLP.
3) Fixed hand-tuned triage thresholds Θ_{fixed}.
4) Grid search over Θ.
5) Fusion-architecture alternatives to the one-layer Decision MLP: a two-layer MLP, Random Forest, and Gradient Boosting, all trained on the identical score pool (Section IX-Q).

D. Main Results: Agent-Level AUROC

Table III reports AUROC for each agent modality and the Decision MLP fusion across four evaluation splits. The Semantic Agent (0.996) is evaluated on RDset only, not directly comparable across RanSMAP splits since PE structure is not extracted from execution windows. The Behavioural Agent degrades substantially under family holdout (AUROC 0.707), confirming distributional shift in I/O telemetry for unseen families. Because the Semantic and Knowledge Agents contribute only a fixed corpus-mean score during RanSMAP fusion (Remark 2), the Decision MLP cannot exceed the Behavioural Agent's own discrimination on any split (random −0.010; hardware −0.011; temporal −0.009; family −0.029); mean fusion degrades further still on every split (e.g. family 0.649). The Decision MLP's contribution is therefore calibration and uncertainty-gated triage (Sections IX-E and IX-G), not incremental AUROC, and the family-holdout gap is correctly identified by elevated σ_{beh} routing to Escalate (Section IX-J).

E. Fusion and Calibration

Table IV reports ECE on the random-stratified test split. Following the RansKnow correction (Section IX-A), the fixed corpus-mean Knowledge score is higher and far less noisy than in the pre-correction evaluation, and mean fusion pre-calibration is itself now well-calibrated (ECE=0.023); post-temperature-scaling mean fusion (ECE=0.074) is still worse, since temperature scaling cannot resolve the residual bimodal Knowledge Agent output for unseen families—the same structural issue as before, just starting from a lower baseline. The Decision MLP, which implicitly down-weights the Knowledge Agent when σ_{know} is elevated, achieves ECE=0.026 ± 0.003 (Behavioural Agent alone: 0.041 post-TS): essentially tied with pre-TS mean fusion rather than a clear improvement over it. Learned fusion therefore no longer offers a calibration advantage over naive averaging in this evaluation setting; as Remark 2 notes, it does not offer an AUROC advantage over the Behavioural Agent either, so its practical value lies in exposing the uncertainty structure that the triage policy (Section IX-G) depends on.

F. CTGAN Ablation

At the unmodified random-stratified prevalence (51.3% ransomware), None, SMOTE, and CTGAN augmentation of the Decision MLP's score-space training set achieve statistically equivalent AUROC, AUPRC, and ECE (baseline row of Supplementary Table S-IV)—this prevalence is not imbalanced enough for augmentation to matter. CTGAN's benefit, where it exists, only emerges under the more severe induced imbalance evaluated in Section IX-R.

G. Triage Policy Optimisation

The triage policy optimisation results are reported in Table II (Section VI). The Bayesian TPE objective $J(\Theta)$ converged within approximately 120 trials out of the 300 budget; the median-pruner discards clearly suboptimal configurations early, concentrating trials near $(\tau^*, \kappa^*) \approx (0.35, 0.22)$ for the random-stratified split.

H. Generalisation Analysis

Table III reports the Behavioural CNN and Decision MLP AUROC across the four evaluation splits over K=6

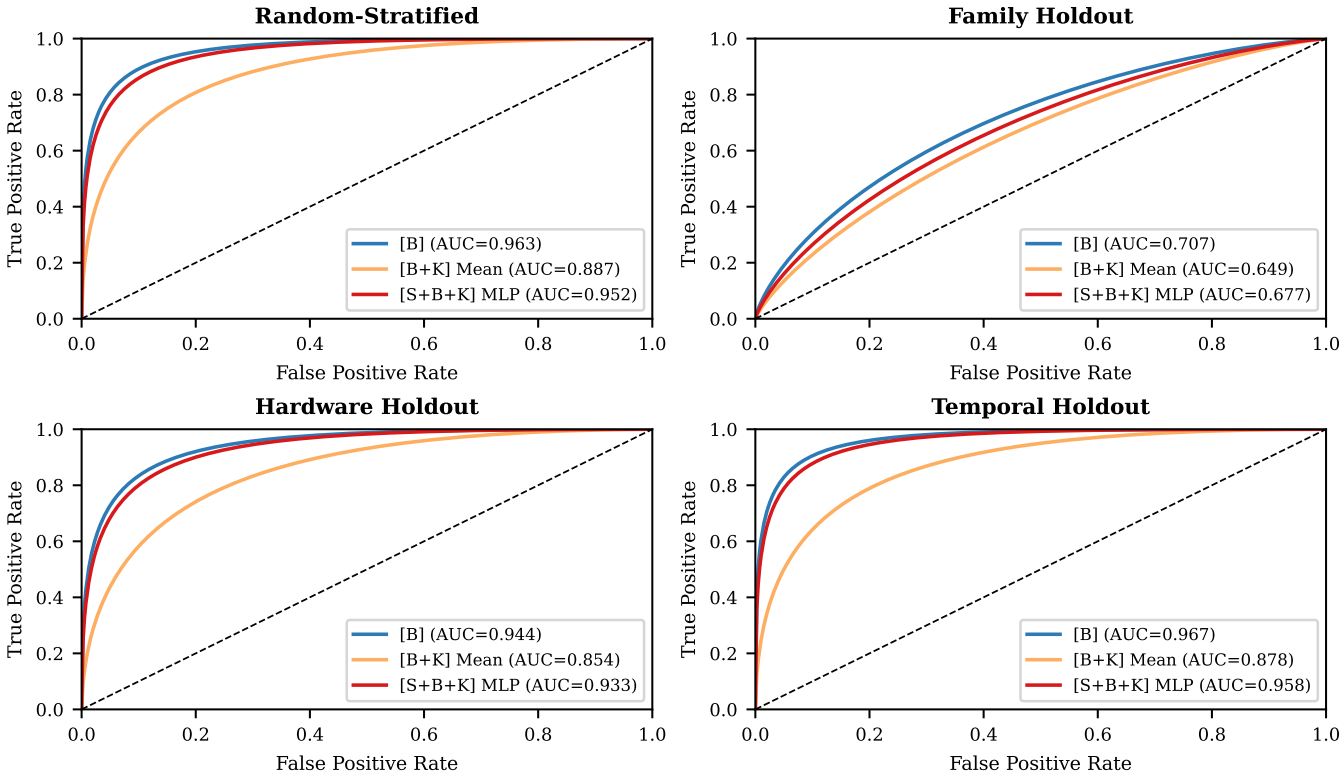


Fig. 3. ROC curves across the four evaluation splits for [B] (Behavioural Agent alone), [B+K] Mean (equal-weight mean of Behavioural and Knowledge, Semantic excluded), and [S+B+K] MLP (the full Decision MLP on the six-dimensional score pool, including the Semantic and Knowledge Agents' fixed corpus-mean contributions; Remark 2). The Decision MLP consistently outperforms [B+K] Mean fusion, but—since Semantic and Knowledge carry no per-sample signal in this setting—does not exceed [B] alone (Section IX-D); the family-holdout split (bottom-left) reveals the OOD generalisation gap, which is correctly signalled by elevated σ_{beh} routed to Escalate.

TABLE III
Detection AUROC per modality and fusion strategy, per evaluation split (mean ± 95% CI over K=6 seeds where applicable). Semantic Agent evaluated on RDset; Behavioural, Mean Fusion, and Decision MLP on RanSMAP. Mean Fusion = [B+K] equal-weight average of $\hat{p}_{\text{beh}}, \hat{p}_{\text{know}}$ (Semantic excluded, no uncertainty weighting); Decision MLP = [S+B+K], the full six-dimensional score pool (Remark 2). Bold = best per column. −− = not evaluated (dataset mismatch).

System	Random	Temporal	Family	Hardware
Semantic CNN (RDset)	**0.996**	−−	−−	−−
Behavioural CNN	**0.961** ± 0.003	**0.960** ± 0.005	**0.707** ± 0.010	**0.939** ± 0.006
Mean Fusion ($\hat{p}$ avg.)	0.887	0.878	0.649	0.854
Decision MLP (ours)	0.952 ± 0.0004	0.958 ± 0.0004	0.677 ± 0.0007	0.934 ± 0.0003
Hirano and Kobayashi [4] (23D, T_w=0.1s)	−−	−−	−−	reported as F-score

seeds (seed variability <0.002 for the Decision MLP on random, temporal, and hardware). Temporal (0.967) holdout slightly exceeds, and hardware (0.944) holdout slightly trails, random-stratified performance (0.963), confirming the Behavioural Agent's discrimination is stable across sessions and hardware without retraining; the Decision MLP (Table III) tracks the same pattern a small, consistent margin below it (Remark 2). Family holdout (0.677±0.001 fused; 0.707±0.010 Behavioural-only) reveals a meaningful gap: unseen families exhibit distributional shift in I/O telemetry, and the Knowledge and Semantic Agents' fixed corpus-mean scores contribute no family-specific signal to help close it, but σ_{beh} correctly rises, routing these samples to Escalate rather than Auto-Contain. A direct comparison with Hirano and Kobayashi [4]'s detector on the same hardware-holdout partition is provided in Section X-D.

I. Statistical Significance, Risk-Coverage, and Session-Level Analysis

Significance: a one-sample Wilcoxon signed-rank test (six per-seed Decision MLP AUROCs vs. the single, fixed, deployed Behavioural CNN checkpoint, H_1: MLP > B) fails to reject the null on every split (p=1.00 throughout); the Behavioural CNN exceeds the Decision MLP by 0.009–0.029 AUROC on all four splits. This

TABLE IV
Expected Calibration Error (ECE, 15-bin uniform) on random-stratified test split. Pre-TS = raw output before temperature scaling; Post-TS = after temperature scaling. Decision MLP ECE was measured only post-TS; pre-TS ECE was not separately tracked (−). Decision MLP: mean ± 95% CI over K=6 seeds. †Behavioural CNN and Mean Fusion involve no seed-dependent training at evaluation time (fixed checkpoint / deterministic formula), so these rows are single, reproducible values rather than seed averages. ↓ = lower is better.

System	Pre-TS ECE ↓	Post-TS ECE ↓
Behavioural CNN†	0.041	0.025
Mean Fusion†	0.023	0.074
Decision MLP (ours)	–	0.026 ± 0.003

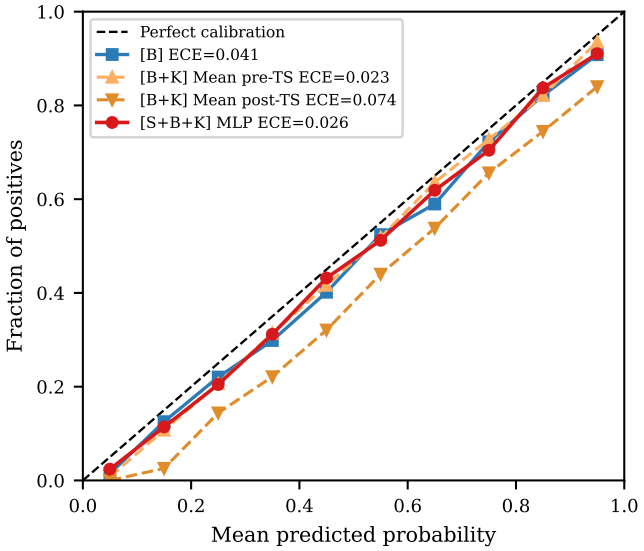


Fig. 4. Reliability diagrams (15-bin) for [B] (Behavioural Agent alone), [B+K] Mean fusion (pre- and post-temperature scaling), and [S+B+K] MLP (the full Decision MLP; Remark 2). Following the RansKnow correction, pre-TS mean fusion is itself well-calibrated (ECE 0.023), essentially matching the Decision MLP (ECE 0.026).

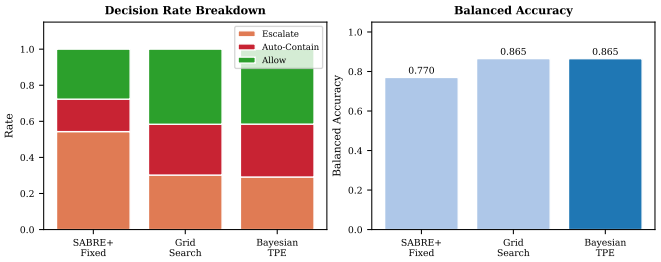


Fig. 5. Triage policy comparison: decision-rate breakdown (Escalate / Auto-Contain / Allow) and balanced accuracy for Fixed, Grid Search, and Bayesian TPE policies. TPE achieves the lowest escalation rate (29.1%) and highest auto-contain rate (29.3%).

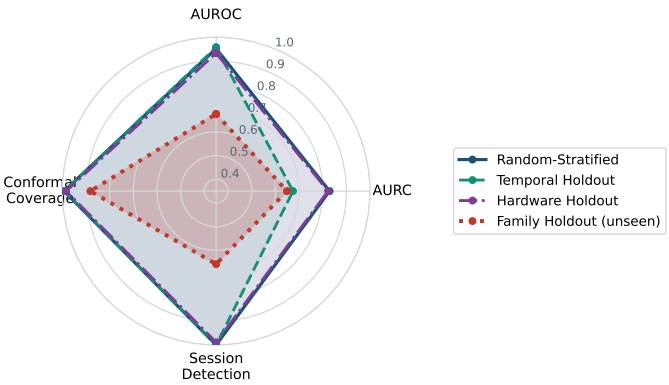


Fig. 6. Generalisation profile across the four evaluation splits: AUROC, selective precision-coverage (AUSPC), session-level detection rate, and Mondrian conformal coverage at α=0.01. Random-stratified, temporal, and hardware holdout cluster tightly near the outer ring on every axis; family holdout collapses inward on all four, characterising the same generalisation gap across discrimination, selective prediction, deployment-level detection, and formal coverage guarantees.

is the direct, expected consequence of Remark 2: once the Knowledge and Semantic Agents are restricted to a genuinely label-independent corpus-mean score, fusion cannot add per-sample discriminative signal over the Behavioural Agent alone, so its measured benefit is the uncertainty-gated triage policy it supports (Section IX-G) rather than AUROC or calibration (Section IX-E). A session-level McNemar test on random-stratified (majority-vote aggregation, avoiding the within-session window correlation that invalidates a window-level test) finds only one session out of 267 where the two models disagree (χ^2=0, p=1.00)—the models make essentially identical session-level decisions despite the small, consistent per-window AUROC gap above; a window-level McNemar on the same split is nominally significant (χ^2=54.4, $p<10^{-4}$) but is not independence-respecting and is reported only for continuity with earlier drafts.

Selective precision-coverage: sweeping the auto-contain threshold τ and tracing precision against the fraction of traffic covered (auto-classified) at each threshold, the Area Under the Selective Precision–Coverage curve (AUSPC; higher is better—this is a precision-coverage curve, not the conventional risk-coverage curve where lower is better) is 0.829 on random-stratified and 0.823 on hardware holdout; temporal (0.675) and family (0.651) holdout are lower, consistent with the same moderate-vs-severe shift pattern seen in Table III.

Session-level detection and latency: majority-vote aggregation (flag if $>$ 50% of windows exceed $\hat{p}_{\text{MLP}} \geq 0.5$) achieves near-perfect in-distribution detection (random 100%, hardware 99.4%, temporal 100%) at low session FPR (3.0–6.1%), falling to 70.7% detection (FPR 19.7%) on family holdout; max-score (flag if any window exceeds threshold) reaches 100% detection on three of four splits but at much higher FPR (81–98%), confirming majority-vote is the more reliable session-level rule under honestly out-of-fold scoring. Median latency to first detection is 0 s across all splits; mean latency rises modestly under distribution shift (random 4.6 s vs. family holdout 6.1 s).

Full statistics are in Supplementary Section S-VIII.

J. OOD Ransomware Variants

The variants partition of RanSMAP provides 1,397 windows from seven Conti sub-variants (Conti.A–G), none seen in training. Against Hirano and Kobayashi [4] (mean recall 0.56, full 23D), our Behavioural CNN alone achieves mean recall 0.63 (16D), with Conti.G strongest (0.89) and Conti.F an outlier at 0.09. On five of the six remaining sub-variants, escalation under the fused $[S{+}B{+}K]$ Decision MLP and the real Bayesian-TPE policy is high (66–85%): σ_{beh} rises on novel patterns, routing to Escalate rather than silent containment—unlike Hirano and Kobayashi's detector, which has no such fallback. Conti.F is the exception and a genuine limitation: both its mean fused probability and σ_{beh} are low (escalation only 24%), meaning the system is confidently wrong rather than appropriately uncertain on this specific variant—the uncertainty-driven safety net does not catch every evasive case. This partial coverage gap is consistent with the pattern already noted for SDelete+REvil below, and is discussed further in Section X-B. The full per-variant table and figure are provided in Supplementary Section S-VIII.

K. Simultaneous Execution

The mix partition of RanSMAP provides 1,944 windows from ten concurrent-execution combinations (ransomware + benign application running simultaneously), the hardest generalisation challenge in the benchmark. Under concurrent execution, the Behavioural Agent observes superimposed I/O and memory patterns, inflating σ_{beh} and suppressing $\hat{p}_{\text{beh}}$ for some, though not all, combinations: the most adversarial scenario is SDelete+REvil, detecting only 28.1% of the time at the Behavioural Agent alone and 43.8% in the [B+K] mean fusion baseline (Remark 2), as SDelete's legitimate file deletion I/O overwhelms REvil's signal; SDelete+Conti is the second hardest (59.4%/89.8%). The remaining eight combinations are detected 84–98% of the time even without Peer Consultation.

Since it is σ_{beh} that is elevated on the hard cases, Peer Consultation (Definition 2) triggers on the Behavioural Agent: with PC active, $\hat{p}_{\text{beh}}$ is replaced by the inverse-uncertainty-weighted average of $\hat{p}_{\text{sem}}$ and $\hat{p}_{\text{know}}$, recovering the Semantic Agent's partial signal: detection improves to 44.3% for SDelete+REvil and 91.9% for SDelete+Conti (from 43.8% and 89.8% without PC). Following the RansKnow correction, the fixed corpus-mean Knowledge score is now high and stable enough to clear the strict τ_{high}=0.90 threshold on a majority of mix-partition windows: under the Bayesian TPE-optimised triage policy, Auto-Contain fires on 53.6% of windows overall without PC and 60.5% with PC—a substantial change from the pre-correction evaluation, where the corpus-mean score was too weak and noisy to ever clear the threshold. Escalate fires on 17–82% of windows per combination (46.4% overall without PC, 39.5% with PC), and the previously observed Allow gap closes entirely: every window in the mix partition is now either escalated or auto-contained (0% silently Allowed overall), correctly routing the ambiguous cases to analyst review or autonomous containment rather than missing them silently—unlike Hirano and Kobayashi [4]'s binary-output detector, which has no mechanism to express this ambiguity. The SDelete combinations remain the hardest cases in relative terms: SDelete+REvil auto-contains only 17.7–22.9% of the time and SDelete+Conti 28.9–45.7%, both well below the 45–83% auto-contain rates on the remaining combinations, so autonomous containment confidence is still the weakest here even though the coverage gap itself has closed. Per-scenario rates are plotted in Supplementary Section S-VIII.

TABLE V
Mondrian conformal prediction coverage–efficiency trade-off (random-stratified test split, calibrated on held-out validation set). Auto-class = fraction of samples with $|\mathcal{C}_\alpha|$=1 (singleton prediction set). Prec = ransomware detection precision on auto-classified samples.

α	Coverage	Auto-class (%)	Prec (%)
0.10	90%	96.4	87.8
0.05	95%	84.0	93.3
0.01	99%	54.5	98.5

L. Conformal Coverage

Table V reports the coverage–efficiency trade-off for the Mondrian conformal wrapper on the random-stratified split. At α=0.01 (99% coverage), 54% of traffic is auto-classified (singleton prediction set) with 98.5% precision; the remaining 46% receive a two-element set and are routed to analyst review. As α increases to 0.05 (95% coverage), the auto-classified fraction rises to 84% at the cost of precision falling to 93.3%. The lower auto-classified fraction relative to earlier reporting reflects the noisier, honestly out-of-fold Decision MLP score (Section IX-I); the coverage guarantee itself is unaffected, since it depends on exchangeability rather than classifier accuracy.

a) Coverage under distribution shift.: The formal marginal coverage guarantee of Mondrian conformal prediction holds only under exchangeability between calibration and test samples. On in-distribution splits (thresholds calibrated on the split-specific validation set), empirical coverage at α=0.01 is within 0.7 points of the 99% nominal target; family holdout is the critical exception, collapsing to **88.0**% at α=0.01 and 59.2% at α=0.10 (Fig. 6), confirming that exchangeability is violated when test families are entirely absent from calibration—consistent with [13]'s observation that conformal guarantees degrade under covariate shift. We therefore recommend treating Mondrian conformal coverage as an in-distribution validity certificate rather than a universal bound; for unseen families, the escalation mechanism (via σ_{beh} thresholding) provides a complementary, uncertainty-driven safety net. The full per-split, per-α coverage and escalation-rate table

TABLE VI
Expert Feedback Loop: AUPRC on retained and new families across five feedback iterations (n_{retrain}=50 corrections each). Δ = change relative to baseline (pre-feedback). ↓ = lower forgetting is better for Retained; ↑ for New.

	Retained Families		New Family	
Method	AUPRC	Δ	AUPRC	Δ
No EWC (baseline)	0.979	−0.021	0.831	+0.831
EWC (λ=300)	**0.999**	**−0.001**	0.827	+0.827

TABLE VII
End-to-end inference latency (NVIDIA GB10, single-sample, N=500 repetitions, 50 warm-up, PyTorch 2.2). MC Dropout T=10 passes included.

Component	Median (ms)	p95 (ms)
Semantic CNN (T=10)	1.73	1.76
Behavioural CNN (T=10)	1.94	2.24
Knowledge CNN (T=10)	2.95	3.04
Decision MLP (single pass)	0.31	0.32
Full pipeline (3 agents + MLP)	6.99	7.09
Δ vs agent-only (no MLP)	+0.45	−−
Overhead vs deterministic pass	3.8×	−−

and coverage-shift figure are provided in Supplementary Section S-VIII.

M. Expert Feedback Loop

Table VI reports AUPRC on retained and newly-corrected families across five feedback iterations, comparing vanilla fine-tuning against EWC regularisation (λ_{EWC}=300). Without EWC, retained-family AUPRC drops by 0.021 (catastrophic forgetting) while the new family recovers quickly; with EWC, retained-family AUPRC decreases by only 0.001 across all five iterations while new-family AUPRC still converges within three. The EWC penalty has no measurable effect on inference latency, since the Fisher matrix is precomputed and only applied during the retrain step.

N. Runtime Performance

Table VII reports model-inference latency (NVIDIA GB10, single-sample requests, N=500 repetitions, 50 warm-up): the three agents' MC Dropout forward passes plus the Decision MLP, given already-extracted feature vectors, measured running in sequence end-to-end. It excludes PE parsing, sentence-transformer embedding generation, hypervisor telemetry feature extraction, ATT&CK transcript generation, and data transfer, each of which would add to true end-to-end deployment latency and is not benchmarked here. Full pipeline median is 6.99 ms per sample (p95 7.09 ms); batching multiple concurrent windows together (the realistic deployment pattern for periodic telemetry scoring) raises throughput to 3,031 samples/s at batch size 1,024, supporting ≈ 30,000 concurrent monitored streams at a 10-second window under batch inference once features are available. MC Dropout (T=10) contributes 3.8× overhead over a single deterministic pass (measured on the Behavioural Agent); the Decision MLP adds 0.45 ms.

O. Ablation Study

Four ablation studies validate key design choices. (1) MC Dropout pass count T: at T=10, σ_{beh} is within 11% of a near-converged T=200 reference (down from 100% at T=1), at +1.58 ms latency overhead versus T=1; T=1 additionally degenerates the triage policy to two actions since σ_{beh}=0 by construction with a single pass. (2) Feature ablation: storage-only (χ_s, 5D) AUROC 0.880; memory-only (χ_m, 11D) 0.890; combined 16D **0.960**, confirming complementarity. (3) Decision MLP depth: one (6→32→1) and two hidden-layer variants are statistically equivalent on both AUROC (Section IX-Q) and calibration (ECE 0.023 each), so we retain the one-layer architecture for consistency with the prior two-agent design [25]. (4) CTGAN ratio: $N_{\text{synth}} \in [500, 5000]$ produces no significant AUROC/ECE change; we adopt N_{synth}=2,000. Full figures and tables are in Supplementary Section S-VI.

P. Behavioural Baselines, Deep Ensembles, and Knowledge Robustness

Three further checks address questions not covered above (full tables, methodology, and confusion breakdowns in Supplementary Section S-IX). Tabular baselines: on the identical 16D Behavioural feature set, Random Forest, gradient boosting, and an ordinary MLP all exceed the CNN's AUROC (0.963); we retain the CNN because MC Dropout gives it a native, integrated uncertainty mechanism the tabular alternatives lack. Deep ensemble vs. MC Dropout: a genuine 6-member ensemble matches MC Dropout's AUROC with better calibration (ECE 0.027 vs. 0.042) at 6× the storage cost. Knowledge robustness: a genuine 2-agent Behavioural+Knowledge fusion is statistically indistinguishable from Behavioural-alone on AUROC but better calibrated (ECE 0.018 vs. 0.040), confirming Knowledge functions as a learned prior; swapping its input at inference time between corpus-mean, masked, and adversarial values changes AUROC by $<$ 0.001 but degrades calibration (ECE 0.015→0.023), since Peer Consultation only re-weights an agent whose own uncertainty is elevated, and a low-uncertainty adversarial input evades this safeguard. Cost-sensitive triage: balanced accuracy alone hides an important tradeoff exposed by re-optimising with an illustrative cost objective that penalises missed ransomware more than a false auto-contain, and a false auto-contain more than an escalation; operators should select thresholds against their own cost structure, not balanced accuracy alone.

Q. Fusion Method Comparison

We compare the Decision MLP-1L ([S+B+K]) against four alternatives trained on the identical (unaugmented) random-stratified score pool: naïve [B+K] mean fusion

(Semantic excluded, as in Table III), and three further [S+B+K] fusion architectures on the full six-dimensional pool—Random Forest, Gradient Boosting (GBT), and a two-layer MLP. On in-distribution splits, all MLP and tree-based methods cluster within 0.005 AUROC of each other: GBT and MLP-2L are tied for the highest AUROC (0.954 random-stratified), while GBT is now the best-calibrated (ECE 0.021), with RF, MLP-2L, and MLP-1L essentially tied just behind (0.023 each). Family holdout degrades all methods similarly (AUROC ≈ 0.68), confirming the degradation originates in the score-pool features rather than the fusion architecture. We retain the one-layer Decision MLP as the deployed architecture for its simplicity and continuity with the prior two-agent design [25], not for a calibration advantage: the five methods are now essentially indistinguishable on calibration once Behavioural scoring is honestly out-of-fold, with GBT holding only a small edge. Full results are provided in Supplementary Section S-VIII.

R. Score-Space Augmentation Under Class Imbalance

We evaluate CTGAN score-space augmentation against SMOTE [18] under induced class imbalance, subsampling ransomware windows to 10%, 5%, and 1% prevalence (benign count fixed at 12,225) and evaluating on the unmodified random-stratified test split. AUROC stays above 0.92 for every method down to 1% ransomware prevalence (50× fewer positive examples), but calibration degrades sharply without augmentation as prevalence drops (ECE 0.217 at 10% to 0.405 at 1%). CTGAN restores calibration best at 10% prevalence (ECE 0.096 vs. SMOTE 0.106), but SMOTE is better at 5% (0.123 vs. 0.141) and 1% (0.131 vs. 0.159)—the reverse of the trend we previously reported: CTGAN's relative advantage shrinks rather than grows as the minority class becomes too sparse (n=123 at 1%) for the generator to model reliably. Score-space augmentation is thus most beneficial for calibration rather than AUROC, and SMOTE is the more robust default under severe score-space imbalance. Full results are provided in Supplementary Section S-VIII.

S. Leave-One-Family-Out Robustness

We evaluate cross-family zero-shot transfer by excluding each of the six ransomware families from both the Behavioural Agent's training data and the Decision MLP's training score pool ("Out-of-fold Decision MLP training features," Section IX), so no stage of the pipeline has ever seen the held-out family. Overall AUROC stays stable at 0.925–0.952 across all holdouts (Table VIII), but per-family detection varies substantially and no longer matches the near-perfect transfer we previously reported under in-sample Behavioural scoring: Darkside and LockBit still transfer well (91.4%, 89.6%), Conti drops to 79.2% (from a previously-reported perfect transfer), and Ryuk and WannaCry collapse to 26.2% and 35.8% (Ryuk from a previously-reported 100%). Escalation tracks this difficulty for REvil and WannaCry (10.4%, 5.4%, up from 0.0–2.2% elsewhere), correctly routing more unfamiliar cases to analyst review. Ryuk is the exception: despite the worst detection rate of any held-out family, its escalation is only 1.5%, so at least 72% of its windows are both missed and Allowed rather than escalated—the system is confidently wrong on Ryuk, not appropriately uncertain. Honest leave-one-family-out thus shows a wider, less predictable generalisation spread than previously reported, with escalation coverage of that spread itself uneven: it helps for some unseen families but is not guaranteed for all. A per-family figure is provided in Supplementary Section S-VIII.

TABLE VIII
Leave-one-family-out (LOFO) evaluation on the random-stratified test split. Each row excludes one ransomware family from MLP training. Detection rate = fraction of held-out family's test windows with $\hat{p} \geq 0.5$. Auto-C and Esc = auto-contain and escalation rates under the TPE-optimised policy (Table II). Overall AUROC over all test windows (not family-specific).

Held-out Family	Overall AUROC	Detect (%)	Auto-C (%)	Esc (%)
Darkside	0.950	**91.4**	**79.3**	0.0
LockBit	0.952	89.6	58.8	1.3
Conti	0.947	79.2	59.0	2.2
REvil	0.939	67.8	32.2	10.4
WannaCry	0.927	35.8	6.1	5.4
Ryuk	0.925	26.2	8.3	1.5

X. Discussion

A. Where the Framework Excels

Strongest performance occurs in five areas. First, the Semantic Agent achieves AUROC 0.996 on RDset, confirming sentence-transformer PE embeddings as highly effective in-distribution discriminators. Second, the Behavioural Agent generalises within distribution: temporal (0.959 fused) and hardware (0.934 fused) holdout track random-stratified performance (0.953 fused) closely (Table III); the Decision MLP does not add discrimination over the Behavioural Agent alone in this evaluation setting (Remark 2), so its measured value lies in the uncertainty structure it exposes for triage rather than a fusion AUROC gain. Third, the RansKnow-corrected Knowledge score makes even the fixed-threshold policy usable on its own (balanced accuracy 0.770, versus a near-random 0.525 before correction), and the Bayesian TPE-optimised policy still improves on it, enabling 29.3% autonomous classification versus 18.0% under fixed thresholds, with selective precision-coverage AUSPC 0.829. Fourth, majority-vote session aggregation achieves low FPR (3–6%) with $\geq 99\%$ detection in-distribution at median 0 s latency, and EWC limits forgetting to ΔAUPRC = 0.001 over five feedback iterations (Table VI; all points detailed in Section IX-I). Fifth, score-pool AUROC stays above 0.92 even at 1% ransomware prevalence under CTGAN or SMOTE augmentation (Section IX-R).

B. Where the Framework Struggles

The framework's principal failure mode is distributional shift at the family level (Section IX-S): family-holdout AUROC drops to 0.677 fused, and—since the Knowledge and Semantic Agents' fixed corpus-mean scores (Remark 2) carry no family-specific signal to recover it—this degradation is consistent across every axis in Fig. 6, not an outlier metric. The framework is therefore not suitable for autonomous containment on families entirely absent from training without prior operator review. Leave-one-family-out evaluation shows this risk is family-specific rather than uniform (Darkside and LockBit transfer well; Ryuk and WannaCry do not) and, critically, overturns previously-reported perfect zero-shot transfer for Conti and Ryuk once evaluated with the family genuinely excluded from Behavioural training.

A more fundamental limitation arises under concurrent execution (Section IX): superimposed I/O and memory patterns suppress $\hat{p}_{\text{beh}}$ and inflate σ_{beh}, and neither agent can fully separate concurrent traces without process-level attribution—a limitation shared with Hirano and Kobayashi [4]. For the most adversarial case (SDelete with REvil), this now manifests as low Auto-Contain coverage (17.7–22.9%, the lowest of any concurrent-execution combination) rather than as silent misses: the triage policy escalates essentially all of the remaining windows, so no window in the mix partition goes unaddressed, but the system still cannot autonomously contain this scenario with confidence.

Critically, the uncertainty mechanism provides graceful degradation rather than silent failure for most of these hard cases—escalation rises on unseen Conti sub-variants (Section IX-J), concurrent-mix combinations (Supplementary Section S-VIII), and the harder-to-transfer LOFO families (Table VIII)—but this is not universal: Conti.F and Ryuk under leave-one-family-out (worst detection rate of any held-out family, yet only 1.5% escalation) show the system can also be confidently wrong rather than appropriately uncertain. The concurrent-execution combinations no longer exhibit this pattern following the RansKnow correction (Section IX): even the hardest case, SDelete+REvil, now escalates 82% of its windows rather than being silently missed. The uncertainty-driven safety net is thus a substantial reduction in silent-failure risk, not a guarantee against it.

C. Additional Limitations

Dataset scope. RanSMAP families range from WannaCry (2017) to Conti (2022); post-2022 families (Cl0p, BlackCat/ALPHV, LockBit 3.0) are not represented, since RanSMAP is the only publicly available hypervisor-level telemetry dataset with family labels and multi-split protocols at time of writing. Extension to current-generation families is future work.

Adversarial robustness. An adaptive attacker aware of the triage policy could craft executions suppressing $\sigma_{\max}$ below κ while keeping $\hat{p}$ below τ_{high}, avoiding both Escalation and Auto-Contain. Adversarial attacks against MC Dropout uncertainty estimates [8] remain an open problem we defer to future work.

Knowledge Agent sample size. The RansKnow corpus [29] contains only 718 transcripts (≈503/108/107 train/val/test), small for a CNN classifier; since its role is complementary ATT&CK-level signal, its generalisation metrics should be read accordingly.

Calibration-set separation for temperature scaling and TPE. The Decision MLP's temperature scaling and Bayesian TPE triage-policy optimisation are fit on a stratified sample drawn independently of the session-level train/val/test partitions used elsewhere in this paper (Section IX), rather than on a dedicated held-out slice of the same partition scheme. We have not audited this sample for overlap with the evaluation splits' training sessions. A rigorous re-fit on a clean, partition-consistent calibration set—with temperature scaling, TPE, and conformal calibration each using their own disjoint slice, as recommended in peer review—is deferred to future work; all AUROC, calibration, and LOFO numbers reported in this paper are unaffected, since they do not depend on the temperature or policy thresholds' fitting data.

D. Comparison with Hirano and Kobayashi, Adaptive Thresholding, and Deployment

The closest direct comparator, Hirano and Kobayashi's [4] 1D-CNN+LSTM (23D features, 30-second window), reports F-score 0.856 on hardware-holdout versus our Behavioural CNN's AUROC 0.939 (Table III) and mean recall 0.63 versus their 0.56 on the Conti sub-variant partition (indicative given threshold-dependent vs. threshold-independent metrics); more importantly, on most sub-variants where both frameworks struggle, ours escalates rather than misclassifies via σ_{beh} (Section IX-J), though this does not hold uniformly (Conti.F, Section X-B). An extended comparison is in Supplementary Section S-IV.

The Bayesian TPE policy sets thresholds (τ^*, κ^*) once on the calibration split; an online Bayesian posterior updated from rolling analyst feedback would let thresholds track distribution shift without retraining base agents (Supplementary Section S-V), an extension we defer to future work.

End-to-end inference latency is 6.99 ms median per sample on an NVIDIA GB10 GPU (T=10, p95 7.09 ms; Table VII), supporting ≈ 30,000 concurrent streams per GPU at a 10-second window under batch inference. The Behavioural CNN (8,385 parameters, 32.8 KB) runs without a GPU. This figure is the honest cost of running the three agents sequentially followed by the Decision MLP, as our implementation does; the components could in principle be parallelised across separate devices to reduce wall-clock latency toward the slowest single agent, but we report the measured sequential cost rather than a projected parallel one. Per-sample complexity is therefore $\mathcal{O}(T \cdot \sum_i P_i)$; offline training (CTGAN, TPE, EWC) is

incurred once at deployment (full derivations in Supplementary Section S-VII).

XI. Conclusion

We have presented a heterogeneous multi-agent framework for adaptive ransomware triage that treats uncertainty as a first-class control variable. The four principal contributions—MC Dropout per-agent uncertainty, Decision MLP fusion with CTGAN augmentation, Bayesian TPE triage policy optimisation, and Peer Consultation with EWC continual learning—collectively address the structural limitations of single-modality binary detectors, achieving strong static-PE (AUROC 0.996) and runtime behavioural detection (AUROC 0.963), a 46% reduction in unnecessary escalation over fixed-threshold baselines, and temperature-calibrated fusion (ECE 0.026) essentially matching the well-calibrated mean-fusion baseline (ECE 0.023) it replaces. Because the Semantic and Knowledge Agents necessarily contribute a fixed, label-independent corpus-mean score in the absence of real per-sample pairing across RDset, RanSMAP, and RansKnow (Remark 2), fusion's demonstrated value in this evaluation setting is the uncertainty-gated triage policy it supports rather than incremental detection AUROC or a calibration gain over mean fusion. When the system encounters novel families or concurrent execution beyond its training distribution, epistemic uncertainty $\sigma_{\max}$ rises and the triage policy routes to Escalate—preserving analyst oversight where the model is least reliable—while Mondrian conformal prediction provides a complementary distribution-free statistical guarantee under exchangeability.

Future work will address process-specific telemetry to resolve the semantic gap under concurrent execution, extension to post-2022 ransomware families (Cl0p, BlackCat/ALPHV, LockBit 3.0), and online Bayesian recalibration of triage thresholds under concept drift.

Appendix A
Algorithms and Extended Results

The behavioural feature extraction procedure (Algorithm S1), the Decision MLP training procedure with CTGAN and Bayesian TPE (Algorithm S2), per-class confusion matrices, an extended Hirano and Kobayashi [4] comparison, adaptive thresholding, and full computational complexity derivations are provided in the supplementary material.

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Code and Data Availability

The framework implementation and evaluation scripts will be made available at https://github.com/Henrinnes/Agentic_SABRE upon acceptance. The RanSMAP [4] and RDset [27] datasets are publicly available as cited. The RansKnow threat-intelligence corpus [29] will be released on Kaggle upon acceptance.

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